

Mini Project 2 — Power BI Dashboard

Sales Performance Dashboard using Power BI

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Dataset:	Superstore Sales Dataset (Kaggle)
Software Used:	Microsoft Power BI Desktop
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Objective

Develop an interactive Power BI dashboard using a real-world sales dataset to analyze business performance and generate meaningful, actionable business insights across regions, product categories, customer segments, and shipping preferences.

Dataset

Source: Superstore Sales Dataset (Kaggle) — as provided in the project brief.

Fields available: Ship Mode, Segment, Country, City, State, Postal Code, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit (13 columns).

Note on scope: This version of the dataset does not include an Order Date or Product Name column. As a result, the required *Monthly Sales Trend* and *Top 10 Products* visuals were substituted with *Sales by Ship Mode* and *Top 10 Cities by Sales* respectively, using fields that were available, in order to fulfil the dashboard's visual and analytical requirements.

Data Preparation

- Dataset imported into Power BI Desktop via Get Data > Text/CSV.
- Checked for missing values and duplicate records in Power Query — none required removal.
- Verified data types for all numeric fields (Sales, Profit, Quantity, Discount) and categorical fields (Region, Category, Sub-Category, Segment, Ship Mode, State, City).
- No date-formatting step was required as no date column was present in this dataset version.

DAX Measures Created

- **Total Sales** = SUM(Sales)
- **Total Profit** = SUM(Profit)

- **Total Quantity** = SUM(Quantity)
- **Sum of Discount** = SUM(Discount)
- **Count of Sales** = COUNT(Sales) — used to build the sales distribution visual

Dashboard Visuals

The dashboard includes 13 visuals — exceeding the required minimum of 10 — providing a comprehensive view of sales and profit performance:

- KPI Cards — Total Profit, Total Quantity, Total Sales
- Sales by Category
- Sales by Sub-Category
- Sales by State
- Sales by City (Top 10, filtered)
- Profit by Region
- Sales by Region (donut chart)
- Sales by Ship Mode
- Customer Segment Analysis (Sales by Segment)
- Discount vs Profit (scatter chart)
- Sales Distribution (count of transactions by sales value)
- Slicers — Region, Category, Segment (for interactive filtering)

Dashboard Screenshot

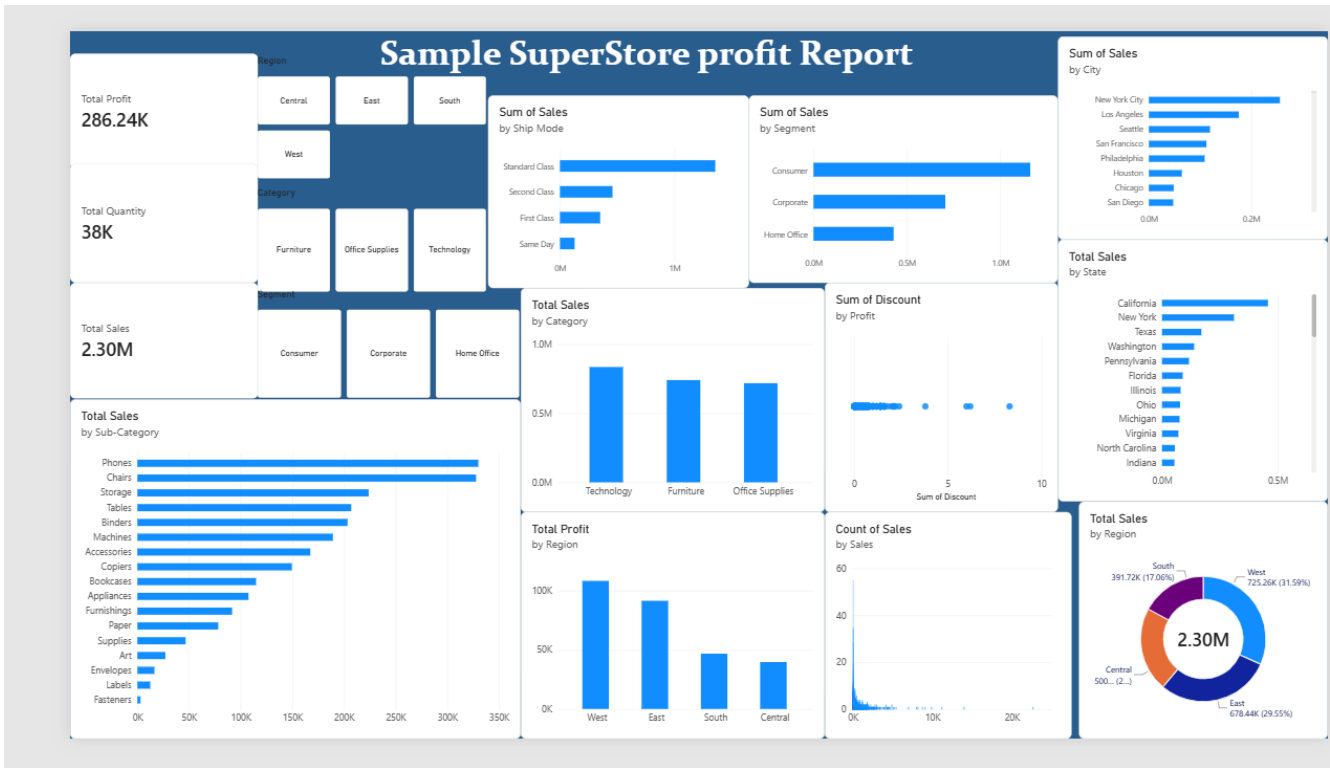


Figure 1: Full Sales Performance Dashboard, showing all KPI cards, visuals, and interactive slicers.

Business Insights & Recommendations

The following insights were derived from the dashboard, along with corresponding business recommendations for each.

1 Sales are heavily concentrated in a few states

California and New York generate a disproportionately large share of total sales compared to most other states, with a long tail of states contributing very little.

Recommendation: Investigate why sales are low in underperforming states — consider targeted regional marketing or promotions to grow presence outside the top states.

2 The West region leads in both sales and profit

The West region accounts for the largest share of sales (31.59%) and the highest profit among all four regions, while Central and South regions trail noticeably behind.

Recommendation: Study what is driving the West region's performance (product mix, customer segment, pricing) and evaluate whether similar strategies can be applied to the Central and South regions.

3 Standard Class is the dominant shipping mode

The majority of orders are shipped via Standard Class, with far fewer using Second Class, First Class, or Same Day options.

Recommendation: Customers appear to prioritize cost over delivery speed. Consider promoting faster shipping options more visibly, or bundling them with incentives, to grow revenue from premium shipping.

4 Consumer segment drives the most sales

The Consumer segment contributes the largest share of total sales, followed by Corporate, with Home Office generating the least.

Recommendation: The Home Office segment represents a growth opportunity — targeted campaigns or product bundles aimed at home-office customers could help expand this segment.

5 Phones and Chairs are the top-performing sub-categories

Phones and Chairs generate substantially higher sales than most other sub-categories, while Envelopes, Art, and Fasteners contribute comparatively little.

Recommendation: Prioritize inventory and marketing spend on top-performing sub-categories, and consider bundling slower-moving items with top sellers to increase their visibility.

6 Higher discounts are associated with lower profit

The discount-vs-profit scatter chart shows that as discount levels increase, profit tends to cluster closer to zero or turn negative, particularly at higher discount percentages.

Recommendation: Review and potentially cap discounting policies on categories where margins are already thin, to avoid discounts eroding profitability.

7 Technology and Furniture rival Office Supplies in sales but likely differ in margin

Sales by Category shows Technology, Furniture, and Office Supplies at broadly similar sales levels, but profit is not evenly distributed the same way.

Recommendation: A follow-up profit-by-category breakdown is recommended to identify which category should be prioritized for growth versus which may need pricing or cost adjustments.

8 Sales values are heavily right-skewed

The sales distribution chart shows the large majority of transactions are low-value, with a small number of high-value orders contributing disproportionately to total sales.

Recommendation: Consider strategies to increase average order value across the many low-value transactions, such as cross-selling, upselling, or minimum-order incentives.

Conclusion

This dashboard provides a comprehensive, interactive view of sales performance across regions, categories, segments, and shipping preferences. The insights above highlight clear opportunities to grow underperforming regions and segments, manage discounting more carefully, and focus resources on top-performing products, while the slicers allow stakeholders to explore the data further on their own.